

TWO-COLOURABLE PLANES OVER NUMBER FIELDS

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ABSTRACT. For a number field F , let $\chi(F^2)$ be the chromatic number of the graph on F^2 in which (x, y) and (x', y') are adjacent when $(x - x')^2 + (y - y')^2 = 1$. We show that $\chi(F^2) = 2$ if and only if some prime of F above 2 ramifies in $F(i)$. That such a prime gives a 2-colouring, by reduction, was known; we prove the converse: if no prime above 2 ramifies in $F(i)$, the graph has an odd cycle. The proof combines Fischer's observation that a 2-colouring is a ring homomorphism with Chevalley's extension theorem for valuation rings. The criterion contains the known cases of real quadratic and real multiquadratic fields and of fields of odd degree, and it holds for the plane of any nondegenerate binary form $x^2 + cy^2$, with $F(\sqrt{-c})$ in place of $F(i)$. We then ask whether, for a real number field, $\chi(F^2)$ is always the least number of colours of a colouring of a single completion F_v^2 that is locally constant (Borel at a real place); the theorem is the case of two colours. For three colours the question predicts $\chi(\mathbb{Q}(\sqrt{d})^2) \geq 4$ for every squarefree $d \equiv 11 \pmod{12}$ (proved since, in a later draft), and for four colours a positive answer for $\mathbb{Q}(\sqrt{167})$ would give a triangle-free 5-chromatic unit-distance graph in the plane.

1. INTRODUCTION

The Hadwiger–Nelson problem asks for the chromatic number $\chi(\mathbb{R}^2)$ of the unit-distance graph of the plane; it is 5, 6 or 7 [5, 7, 21]. For a subfield $F \subset \mathbb{R}$ the plane F^2 is a subgraph, and since the adjacency $(x - x')^2 + (y - y')^2 = 1$ is algebraic, $\chi(F^2)$ makes sense for every field F . Woodall showed $\chi(\mathbb{Q}^2) = 2$ [24]. For real quadratic fields, $\chi(\mathbb{Q}(\sqrt{N})^2) = 2$ exactly when $N \not\equiv 3 \pmod{4}$: Johnson proved that these planes are 2-colourable [14] (we know his paper through [9, 19]), and Fischer proved both directions [9, Theorem 8]. Moorhouse gave proofs of the 2-colourable cases and showed $\chi(F^2) = 2$ for every field of odd degree [18, Theorem 7.1, Lemma 8.4, Theorem 8.5].

The colourings of Woodall, Fischer and Moorhouse are reductions modulo a prime above 2. Madore's reduction argument [16, Proposition 3.2, ¶6.6] needs $x^2 + y^2$ to stay anisotropic modulo the prime, which never happens above 2, where $x^2 + y^2 \equiv (x + y)^2$. For primes above 2 he gave a version modulo the square of the prime, which applies when a prime above 2 is unramified over $\mathbb{Q}$ [16, ¶3.7, Proposition 3.8], and a separate argument for $\mathbb{Q}(\sqrt{2})$ [16, Proposition 3.9]. Writing points as $z = x + iy \in F(i)$, a public repository of July 2026 [17] proved the general statement: if a prime of F above 2 ramifies in $F(i)$, reduction modulo the prime above it gives a 2-colouring [17, Theorem A']. We recall its short proof as Proposition 3, in the slightly stronger form of a locally constant colouring of the completion; the argument is the same.

The other direction, an odd cycle whenever no prime above 2 ramifies in $F(i)$, had been proved field by field. For real quadratic fields Fischer proved it with the ring homomorphism attached to a 2-colouring [9, Theorems 1, 7 and 8], and Moorhouse, for $\mathbb{Q}(\sqrt{p})$ with p prime, with a cycle from Pell's equation [18, Theorem 8.6]; it follows for every field with such a quadratic subfield. We prove it for every number field, by combining Fischer's ring homomorphism with Chevalley's extension theorem.

Theorem 1. *Let F be a number field. Then $\chi(F^2) = 2$ if and only if some prime of F above 2 ramifies in $F(i)$.*

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These results were obtained with the assistance of an AI system (Claude, by Anthropic), which carried out the computations and the checks and helped to draft this note. Two further sessions of the same system, given only the manuscript, refereed it; no mathematician has checked it yet. The author is not a professional mathematician and submits it for independent verification.

The *only if* direction, Proposition 4, is the new part. We have not found it in the literature we could consult, but we have not seen Johnson's paper [14], his survey [15] or the paper of Benda and Perles [2]. Section 2 proves the theorem, Section 3 draws the consequences, and Section 4 puts it in a local-global frame.

2. PROOF

Let $L = F[t]/(t^2 + 1)$; it is the field $F(i)$ if $i \notin F$, and $F \times F$ if $i \in F$. Identify F^2 with L by $(x, y) \mapsto x + ty$; then $x^2 + y^2$ is the norm $N(z) = z\bar{z}$, and the unit vectors are

$$T = \{z \in L : z\bar{z} = 1\}.$$

T is a group under multiplication, so its additive span $A = \mathbb{Z}[T]$ is a subring of L . The graph is the Cayley graph of the additive group L with connection set T , so it is a disjoint union of translates of A , each a copy of the Cayley graph $\text{Cay}(A, T)$. Every $u \in T$ is a unit of A , since $u^{-1} = \bar{u} \in T$.

Lemma 2 (Fischer [9, Theorem 1(iii)], for $F \subset \mathbb{R}$). $\chi(F^2) = 2$ if and only if there is a ring homomorphism $\varphi : A \rightarrow \mathbb{F}_2$.

Proof. Since $-1 \in T$, every element of A is a sum of unit vectors. If $\text{Cay}(A, T)$ is bipartite, let $\varphi(a)$ be the parity of the length of a walk from 0 to a ; it is well defined and additive. It is also multiplicative: if a is a sum of m unit vectors and b a sum of n , then ab is a sum of mn products of unit vectors, which are unit vectors. Conversely, a ring homomorphism sends each $u \in T$, a unit of A , to $1 \in \mathbb{F}_2^\times$, so $\varphi(a + u) = \varphi(a) + 1$ and φ is a proper 2-colouring of $\text{Cay}(A, T)$, which extends to the translates. (The graph has edges, so $\chi \geq 2$.) $\square$

Proposition 3 ([17, Theorem A']; cf. [16, Propositions 3.2, 3.8 and 3.9]). *If a prime v of F above 2 ramifies in L , then F_v^2 , and hence F^2 , has a 2-colouring that is locally constant for the topology of F_v .*

Proof. Let w be the place of L above v . It is the only one, so conjugation preserves $|\cdot|_w$; a unit vector u of $F_v^2 = L_w$ therefore satisfies $|u|_w^2 = |u\bar{u}|_w = 1$, so $u \in \mathcal{O}_w^\times$. Since L_w/F_v is ramified, the residue fields of w and v agree and conjugation acts trivially on them; reducing $u\bar{u} = 1$ gives $u^2 \equiv 1 \pmod{\mathfrak{m}_w}$, and in characteristic 2 this is $u \equiv 1 \pmod{\mathfrak{m}_w}$. Choose an additive map λ from the residue field to $\mathbb{F}_2$ with $\lambda(1) = 1$. Unit steps preserve the cosets of $\mathcal{O}_w$ in L_w ; fix a representative r of each coset and colour z by $\lambda((z - r) \bmod \mathfrak{m}_w)$. A unit step changes the colour. The colouring is constant on the cosets of $\mathfrak{m}_w$. $\square$

Proposition 4. *If there is a ring homomorphism $\varphi : A \rightarrow \mathbb{F}_2$, then some prime of F above 2 ramifies in L .*

Proof. Suppose first that $i \in F$, so $L = F \times F$ and $T = \{(s, 1/s) : s \in F^\times\}$. For $s \neq 0, 1$ the sum of $(s, 1/s)$ and $(1 - s, 1/(1 - s))$ is $(1, 1/(s(1 - s)))$; the difference of the sums for $s = 2$ and $s = 3$ is $(0, -1/3) \in A$. The set $\{x \in F : (0, x) \in A\}$ is an ideal of the second projection of A , which contains $F^\times$ and is therefore F ; this ideal is non-zero, so $A \supseteq 0 \times F$, by symmetry $A = F \times F$, and $F \times F$ has no homomorphism to $\mathbb{F}_2$ because 2 is invertible. So $i \notin F$ and L is a field.

By Zorn's lemma the pairs (B, ψ) of a subring $B \supseteq A$ of L and a homomorphism ψ from B to an algebraic closure of $\mathbb{F}_2$ extending φ have a maximal element (V, ψ) , and by Chevalley's theorem [1, Lemma 5.19 and Theorem 5.21] V is a valuation ring of L with maximal ideal $\ker \psi$, which meets A in $\ker \varphi$. As V contains $\mathbb{Z}$ and is integrally closed, $V \supseteq \mathcal{O}_L$; as $2 \in \ker \varphi$, $V \neq L$. So V is the valuation ring of a place w of L above 2, and every $u \in T$ satisfies $u \equiv 1 \pmod{\mathfrak{m}_w}$. Let v be the place of F below w .

The map $p \mapsto (p + t)/(p - t)$ is a homeomorphism from F_v minus the roots of $p^2 + 1$ onto $T(F_v) \setminus \{1\}$, where $T(F_v)$ is the group of norm-one elements of $L \otimes_F F_v$, and it maps F onto $T \setminus \{1\}$; so T is dense in $T(F_v)$.

- If v splits in L , then $L \otimes_F F_v = F_v \times F_v$ and $z \mapsto z_w$ identifies $T(F_v)$ with $F_v^\times$, which is unbounded; so some element of T is not integral at w , a contradiction.

- If v is inert in L , with residue field $\mathbb{F}_q$, then conjugation acts on the residue field $\mathbb{F}_{q^2}$ of w as $x \mapsto x^q$, and the unit vectors $y/\bar{y}$ with $y \in \mathcal{O}_w^\times$ reduce to y^{1-q} , which runs through all $q+1 \geq 3$ roots of $x^{q+1} = 1$. Each residue class is open, so some element of T is not $\equiv 1 \pmod{\mathfrak{m}_w}$, a contradiction.

Hence v ramifies in L . □

Proof of Theorem 1. Combine Lemma 2 with Propositions 3 and 4. □

3. CONSEQUENCES

For a multiquadratic field the condition is a computation in the group $\mathbb{Q}_2^\times/\mathbb{Q}_2^{\times 2}$, of order 8, generated by -1 , 5 and 2 , and written multiplicatively. For a subgroup H let $\mathbb{Q}_2(\sqrt{H})$ be the field generated over $\mathbb{Q}_2$ by square roots of representatives of the elements of H ; then $\mathbb{Q}_2(\sqrt{5})$ is the only unramified quadratic extension of $\mathbb{Q}_2$.

Corollary 5. *Let $F = \mathbb{Q}(\sqrt{a_1}, \dots, \sqrt{a_k})$ with nonzero integers a_j . Then $\chi(F^2) = 2$ unless some quadratic subfield $\mathbb{Q}(\sqrt{c})$, with $c \neq 1$ a squarefree integer, has $c \equiv 3 \pmod{4}$; that is, F^2 is 2-colourable exactly when the plane of every quadratic subfield is.*

For real F this is not new: the *if* half is [17, Corollary B], and the *only if* half follows from Fischer's theorem, since F^2 contains the plane of each subfield. For non-real F the same reasoning works with Theorem 1 in place of those results. We include the corollary to show how Theorem 1 reads in this case.

Proof. The primes of F above 2 are conjugate. Let G be the image of $\langle a_1, \dots, a_k \rangle$ in $\mathbb{Q}_2^\times/\mathbb{Q}_2^{\times 2}$ and $G' = G\langle -1 \rangle$, so that $F_v = \mathbb{Q}_2(\sqrt{G})$ and $F_v(i) = \mathbb{Q}_2(\sqrt{G'})$. For a subgroup H the ramification index of $\mathbb{Q}_2(\sqrt{H})/\mathbb{Q}_2$ is $|H|/|H \cap \langle 5 \rangle|$. If $-1 \in G$ the extension $F_v(i)/F_v$ is trivial. Otherwise $|G'| = 2|G|$, and $F_v(i)/F_v$ is unramified exactly when $|G' \cap \langle 5 \rangle| = 2|G \cap \langle 5 \rangle|$, that is, when $5 \in G' \setminus G$, that is, when $-5 \in G$. So no prime above 2 ramifies exactly when G contains -1 or -5 , that is, when the squarefree representative $c \neq 1$ of some element of the group generated by the a_j modulo squares is $\equiv 7$ or $\equiv 3 \pmod{8}$. These c are those of the quadratic subfields. □

For real quadratic fields Theorem 1 is Fischer's theorem [9, Theorem 8]. For F of odd degree, some prime v above 2 has odd degree over $\mathbb{Q}_2$; then $i \notin F_v$, and since $\mathbb{Q}_2(i)/\mathbb{Q}_2$ has ramification index 2 and $F_v/\mathbb{Q}_2$ an odd one, $F_v(i)/F_v$ is ramified, which is Moorhouse's Theorem 7.1 [18].

Remark 6. Nothing in the proof uses the particular form $x^2 + y^2$. Let $c \in F^\times$ and $q = x^2 + cy^2$. With $L = F[t]/(t^2 + c)$ and $N(x + ty) = x^2 + cy^2$, the same argument shows that the graph $q(x - x', y - y') = 1$ on F^2 is bipartite if and only if some prime of F above 2 ramifies in $F(\sqrt{-c})$. Every nondegenerate binary form that represents 1 is equivalent to such a form, and a form that does not represent 1 gives a graph without edges. Nondegeneracy is needed: for $q = x^2$ the components are the sets $(x_0 + \mathbb{Z}) \times F$, and the graph is bipartite over every F .

Example 7. Theorem 1 also decides fields that are not multiquadratic. Let F be generated by a root of $\alpha^4 - 2\alpha^3 - 2\alpha - 3$ (Galois group S_4 , so that F has no quadratic subfield), of $\alpha^4 - 6\alpha^2 + 4\alpha + 2$ (whose only quadratic subfield is $\mathbb{Q}(\sqrt{2})$) or of $\alpha^4 - 5\alpha^2 - 7$ (whose only quadratic subfield is $\mathbb{Q}(\sqrt{53})$). In each case no prime above 2 ramifies in $F(i)$, so F^2 has an odd cycle, although the planes of $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{53})$ are 2-colourable. Exact integer relations $\sum_k c_k u_k = 0$ among unit vectors $u_k = z_k/\bar{z}_k$ with $\sum_k c_k$ odd, which are closed walks of odd length $\sum_k |c_k|$, are found in these three fields by the script `odd_walks.gp` of [10]; for three fields in which a prime above 2 ramifies, the same search finds only relations with even sum. The script `two_colour_criterion.gp` checks the criterion against Fischer's theorem for the 242 squarefree N with $2 \leq N \leq 400$, against Moorhouse's theorem for nine fields of degree 3, 5 and 7, and against Corollary 5 for 1522 biquadratic fields.

4. A LOCAL–GLOBAL QUESTION

Call a number field *real* if it has a real place. For a place v of a real number field F , let $\chi_{\text{loc}}(v)$ be the least number of colours of a proper colouring of F_v^2 that is locally constant or, at an archimedean place, Borel. At a real place this is the Borel chromatic number $\chi_{\text{B}}(\mathbb{R}^2)$, which is at least 5 by Falconer’s theorem [8] and at most 7 by the hexagonal colouring. At a complex place, and at a finite place where -1 is a square in F_v , the form $x^2 + y^2$ is isotropic and even the ordinary chromatic number of F_v^2 is infinite, by a theorem of Davies [3] (see [12, Proposition 3]); there $\chi_{\text{loc}}(v) = \infty$. At a finite place where $x^2 + y^2$ is anisotropic, the unit vectors are integral, and $\chi_{\text{loc}}(v)$ is the least chromatic number of the finite graphs $\text{Cay}(\mathcal{O}_w/\mathfrak{m}_w^r, T_r)$, where w is the place of $F(i)$ above v and T_r is the image of the norm-one units: unit steps preserve the cosets of $\mathcal{O}_w$, which can be coloured independently, and a locally constant colouring of the compact set $\mathcal{O}_w$ is constant on the cosets of some $\mathfrak{m}_w^r$. Restriction gives

$$\chi(F^2) \leq \min_v \chi_{\text{loc}}(v),$$

and every upper bound below 7 that we know for a plane over a number field is of this kind: reductions at one finite place [9, 16, 17, 18].

Question 8. Is $\chi(F^2) = \min_v \chi_{\text{loc}}(v)$ for every real number field F ?

Theorem 1 answers Question 8 whenever one side is 2: if $\chi(F^2) = 2$, Lemma 2 and Proposition 4 give a prime v above 2 that ramifies in $F(i)$, and Proposition 3 gives $\chi_{\text{loc}}(v) = 2$.

Two facts bound the local side from below. First, if v lies above a prime p , then $F_v \supseteq \mathbb{Q}_p$, and a locally constant colouring of F_v^2 restricts to a locally constant, hence measurable, colouring of $\mathbb{Q}_p^2$; so $\chi_{\text{loc}}(v) \geq \chi_{\text{M}}(\mathbb{Q}_p^2)$, the measurable chromatic number. For $p \equiv 3 \pmod{4}$, the ratio bound of Delsarte and Hoffman on the compact group $\mathbb{Z}_p[i]$ gives $\chi_{\text{M}}(\mathbb{Q}_p^2) \geq 4$ for $p \geq 11$, $\chi_{\text{M}}(\mathbb{Q}_p^2) \geq 5$ for $p \geq 23$, and $\chi_{\text{M}}(\mathbb{Q}_p^2) \geq 7$ for $p = 71$ and for $p \geq 103$ [12, Theorem 1 and Table 1]. Second, if a field K embeds in F_v , then $\chi_{\text{loc}}(v) \geq \chi(K^2)$.

Three colours. For squarefree $d \geq 2$, the field $\mathbb{Q}(\sqrt{d})$ has a place with $\chi_{\text{loc}}(v) \leq 3$ exactly when $d \not\equiv 11 \pmod{12}$. If $d \not\equiv 3 \pmod{4}$, Proposition 3 gives two colours at 2. If $d \not\equiv 2 \pmod{3}$, a place v above 3 has residue field $\mathbb{F}_3$, the place above it has residue field $\mathbb{F}_9$, and the norm-one units reduce to $\{\pm 1, \pm i\}$; so $a + bi \mapsto a + b \pmod{3}$ colours the first level with three colours (cf. [16, Proposition 3.2], [9, Theorem 9]). Conversely, let $d \equiv 11 \pmod{12}$. The real places have $\chi_{\text{loc}} \geq 5$. The prime 3 is inert, and the places above 3, the places above primes $p \equiv 1 \pmod{4}$, and the place above 2 when $d \equiv 7 \pmod{8}$ contain i . When $d \equiv 3 \pmod{8}$, the place above 2 has $F_v = \mathbb{Q}_2(\sqrt{3})$, which contains $\sqrt{11}$ because 33 is a square in $\mathbb{Q}_2$; and a place above 7 has $F_v \supseteq \mathbb{Q}_7$, which contains $\sqrt{11}$ because $11 \equiv 2^2 \pmod{7}$. In both cases $\chi_{\text{loc}}(v) \geq \chi(\mathbb{Q}(\sqrt{11})^2) = 4$ [13]. Above a prime $p \equiv 3 \pmod{4}$ with $p \geq 11$, $\chi_{\text{loc}}(v) \geq \chi_{\text{M}}(\mathbb{Q}_p^2) \geq 4$. So Question 8 predicts $\chi(\mathbb{Q}(\sqrt{d})^2) \geq 4$ for every squarefree $d \equiv 11 \pmod{12}$. This has been checked, by unit-distance graphs with certified proofs of non-3-colourability, for 27 values of d [13]; no counterexample is known. [Added 3 October 2026.] A later draft [11] proves $\chi(\mathbb{Q}(\sqrt{d})^2) \geq 4$ for every $d \equiv 11 \pmod{12}$. So for three colours the answer to Question 8 is positive for every real quadratic field: $\chi(\mathbb{Q}(\sqrt{d})^2) \leq 3$ exactly when a place above 2 or 3 has a locally constant 3-colouring.

Colourings that are locally constant at several anisotropic finite places colour a categorical product of finite level graphs, since T is dense in the product of the local groups $T(F_v)$; they can do better than every single place only through a counterexample to Hedetniemi’s conjecture, as Moorhouse observed [18, §9]. Such counterexamples exist with many colours [20], so for large numbers of colours Question 8 is a genuine restriction to single places; with three colours they do not exist [6].

Four colours. Call a number field F *admissible* if $\sqrt{3} \notin F$ and $i \in F_v$ for every place v of F above 2, 3, 7, 11 and 19. A real admissible field has $\chi_{\text{loc}}(v) \geq 5$ at every place v : $\chi_{\text{loc}} \geq 5$ at the real places; $\chi_{\text{loc}} = \infty$ at the complex places, at the places above 2, 3, 7, 11 and 19 and at those above primes $p \equiv 1 \pmod{4}$; and at a place above a prime $p \equiv 3 \pmod{4}$ with $p \geq 23$, $\chi_{\text{loc}}(v) \geq \chi_{\text{M}}(\mathbb{Q}_p^2) \geq 5$. The margins are small: the ratio bound on the density of a colour class is 0.24908 at $p = 23$ and 0.2499973 at $p = 31$, against $1/4$ [12, Table 1]. An admissible field

also has no unit triangle, since the third vertex of a unit equilateral triangle on 0 and (a, b) is $(\frac{a}{2} - \frac{\sqrt{3}}{2}b, \frac{b}{2} + \frac{\sqrt{3}}{2}a)$. The smallest admissible real quadratic field is $\mathbb{Q}(\sqrt{167})$; the next are $d = 887, 1055, 1319$ and 1823 . Fischer had singled out the same field: $\mathbb{Q}(\sqrt{167})^2$ has no additive k -colouring for $k < 11$ [9, Example 4]. So a positive answer to Question 8 for $\mathbb{Q}(\sqrt{167})$ would give $\chi(\mathbb{Q}(\sqrt{167})^2) \geq 5$, and by the theorem of de Bruijn and Erdős [4] a finite triangle-free 5-chromatic unit-distance graph in the plane. According to the abstract of [22, Chapter 55], Soifer asked for such graphs in a problem paper of 2019; according to [23], de Grey has found one in $\mathbb{R}^3$. We have not found a planar one in the literature we could consult. The searches of [10] have not found one over $\mathbb{Q}(\sqrt{167})$: its unit vectors have large denominators, and the graphs grown there stay sparse.

The real place. If Question 8 had a positive answer for every real quadratic field, then $\chi(\mathbb{R}^2) = \chi_B(\mathbb{R}^2)$. Take a prime $d \equiv 7 \pmod{8}$ that is a quadratic non-residue modulo every prime $p \equiv 3 \pmod{4}$ below 100 other than 71; the smallest is $d = 186\,023$. In $\mathbb{Q}(\sqrt{d})$ the places above 2, above the primes $p \equiv 1 \pmod{4}$ and above those primes $p \equiv 3 \pmod{4}$ contain i , and every other finite place lies above 71 or above a prime $p \geq 103$, so $\chi_{\text{loc}}(v) \geq \chi_M(\mathbb{Q}_p^2) \geq 7$. The question would then give $\chi(\mathbb{Q}(\sqrt{d})^2) = \chi_B(\mathbb{R}^2)$, while $\chi(\mathbb{Q}(\sqrt{d})^2) \leq \chi(\mathbb{R}^2) \leq \chi_B(\mathbb{R}^2)$. Whether $\chi(\mathbb{R}^2) = \chi_B(\mathbb{R}^2)$ is open, so this part of the question is the most speculative.

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